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 - a) Using the CEA software, calculate and plot the combustion temperature (also known as the adiabatic flame temperature) of a $\text{CH}_4\text{--O}_2$ gaseous rocket thrust chamber with r changing from 1 to 10. The combustion chamber pressure is 68 atm and flow fully expands to ambient at 1 atm. The reactants enter it at 298.15 K.
 - b) Plot also I_{sp} , A_e/A_t , C^* , C_τ (C_f in CEA), exit Mach M_e , M (molar mass of the product) with r .
 - c) Compute $r_{\text{stoichiometric}}$ for ideal reaction and show that I_{sp} maximum does not correspond to $r_{\text{stoichiometric}}$.
- 2- We have a nozzle with an inlet-to-throat area ratio of 4. For $\gamma=1.1, 1.2, 1.3, 1.4, 1.5$, and 1.6 , plot P_{02}/P_{01} and P_2/P_1 vs γ . (1: Inlet of combustor, 2: exit of combustor)